Kai Yui Man (Karl)

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Profile

Completed MSc studies in Robotics and Artificial Intelligence at UCL (final award pending), following a First-Class BSc in Computer Science and Artificial Intelligence.

Education

University College London (UCL) Sept 2025 – Sept 2026
MSc Robotics and Artificial Intelligence *London, UK*
Taught-module average: 75.2/100; final classification pending.
Dissertation: Force-Conditioned Flow Matching for Contact-Aware Robot Action Generation.

University of Sussex Sept 2022 – May 2025
BSc Computer Science and Artificial Intelligence, First-Class Honours (72/100) *Brighton, UK*

Research Experience

Force-Conditioned Flow Matching for Contact-Aware Robot Action Generation Jan 2026 – Present
MSc Dissertation & Real-Robot Research Extension, UCL; Supervisor: Dr Valerio Modugno

- Developed flow-matching action policies integrating vision, proprioception, and temporal force observations for contact-rich manipulation.
- Compared force-conditioning designs (AdaLN, cross-attention, prefix–suffix conditioning, and dual-path force attention) under controlled simulation experiments.
- Found that force feedback improved simulated task performance across weight discrimination, door manipulation, and precision insertion, with particularly strong gains on contact-sensitive tasks.
- Built PyTorch/MuJoCo/robosuite training and closed-loop evaluation pipelines; currently extending the study to a Franka arm using joint-torque-derived force estimates.

British Sign Language Recognition and Learning System Sept 2024 – May 2025
BSc Dissertation, University of Sussex
Developed a real-time YOLO-based sign-language recognition system, compared model variants, and integrated the model into an interactive web application.

Professional and Teaching Experience

AI Computer Vision Engineer – Appraiseye June 2026 – Present
Cambridge, UK

- Develop computer vision pipelines for biomedical and surgical video analysis, including segmentation, tracking, and temporal analytics.
- Prototyped an interactive operating-room 3D digital twin using Gaussian Splatting, COLMAP, and LiDAR-derived geometry; developed automated annotation workflows.

Postgraduate Teaching Assistant – University College London Oct 2026 – Present
Support postgraduate laboratory sessions.

Digital & IT Summer Intern – Cathay Pacific Jul – Aug 2024
Contributed data-processing and software components to an internal cross-functional technology project.

Selected Project

Reinforcement Learning for Robotic Arm Reaching Apr – May 2024
Trained and evaluated a simulated arm-reaching policy using Panda-Gym and Stable-Baselines3, exploring continuous control and reward design.

Technical Skills

Programming:	Python, C++, Linux, Git
Robot Learning:	PyTorch, flow matching, imitation learning, reinforcement learning, generative action policies
Robotics:	MuJoCo, robosuite, ROS 2, Franka, robot manipulation, control and estimation
Large Model:	LLM, VLM, World Model
Computer Vision:	OpenCV, YOLO, SAM, DINOv2, COLMAP